

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Data Interpretation

Code of Conduct:

I P.Jyothika(192311156(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	

Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

ANSWER:

1. Real-Time System Design Trade-off

Method	Speed	Accuracy	Suitability
Viola-Jones	Very fast	78%	Good for strict

			real-time
YOLO	Moderate	92%	Best balance
Deep Learning	Slow	95%	Better for accuracy, not latency

Answer: YOLO should be selected.

Viola-Jones is faster but has lower accuracy. Deep Learning gives the highest accuracy but is too slow for strict latency requirements. YOLO provides **92% accuracy with moderate speed**, giving a good balance between accuracy and real-time performance. It is suitable for surveillance systems where quick detection and reliable results are both important.

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method Normal Occlusion Low Light

Viola-Jones 80% 60% 55%

YOLO 90% 75% 70%

Deep Learning 95% 85% 80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

ANSWER:

Method	Normal	Occlusion	Low Light	Overall
Viola-Jones	80%	60%	55%	Low
YOLO	90%	75%	70%	Good
Deep Learning	95%	85%	80%	Best

Answer: Deep Learning provides the best overall performance.

It has the highest accuracy in normal, occluded, and low-light conditions. Viola-Jones performs poorly when lighting and visibility change. YOLO is more robust than Viola-Jones, but Deep Learning performs best across all environments. The main trade-off is its higher computational cost.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions

- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

ANSWER:

Method A:

- Produces stable and reliable motion vectors.
- Less sensitive to noise.
- May miss small or fine movements.
- Suitable when stability is more important.

Method B:

- Detects small and fine movements.
- Produces noisy vectors.
- May require filtering or noise reduction.
- Suitable when detecting detailed motion is important.

Answer: Method A is more suitable for general motion tracking in dynamic scenes because stable vectors provide more reliable tracking. However, if the application needs to detect very small movements, Method B can be selected with a noise-filtering technique.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

ANSWER:

GMM performs differently depending on the background:

- In **dynamic backgrounds**, moving trees, water, or changing objects can cause many false positives.
- In **static backgrounds**, it may miss some objects if the background model is not updated properly.
- Therefore, GMM alone is **not sufficient** for reliable detection.
- Adaptive background updating can reduce missed detections.
- Combining GMM with **morphological filtering, shadow removal, or object detection methods such as YOLO** can improve performance.

Answer: GMM is useful for background subtraction, but it should be combined with other

techniques to handle complex real-world environments

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

ANSWER:

Method	Error	Computation	Best Use
Method A	5%	High	Offline
Method B	10%	Low	Real-time

Answer:

For real-time applications, Method B is more suitable because it requires less computation and can provide faster results, even though its error is slightly higher.

For offline applications, Method A is better because its lower error of 5% gives more accurate results, and higher computational cost is acceptable when processing time is not critical.

Conclusion: Choose Method B for real-time systems and Method A for offline/high-accuracy systems.